Graph Traversal

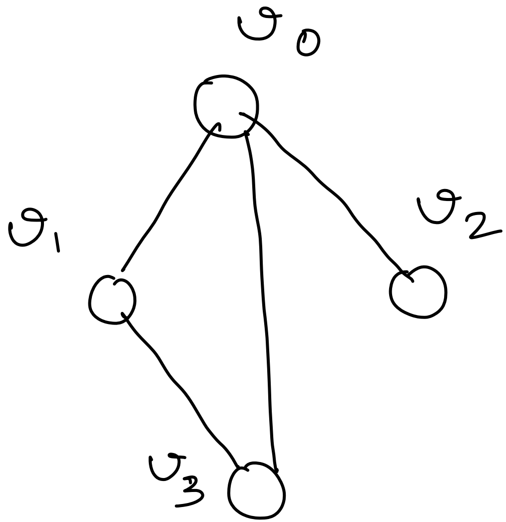
Depth First (DFS)

Breadth First (BFS)

Tree is a special type of graph

- Tree do not have cycle.
- All nodes are connected in a tree.

BFS



adjMatrix

	v_0	v_1	v_2	v_3
v_0	0	1	1	1
v_1	1	0	0	1
v_2	1	0	0	0
v_3	1	1	0	0

v_0	v_1	v_2	v_3
~~F~~ T	~~F~~ T	~~F~~ T	~~F~~ T

isVisited

queue

~~v_0~~	~~v_1~~	~~v_2~~	~~v_3~~	~~v_3~~

O/p : v_0 v_1 v_2 v_3

Current → ~~v_0~~ ~~v_1~~ ~~v_2~~ ~~v_3~~ v_3

BFS()
- Create isVisited array of size vertexCount.
- Set all elements in isVisited to FALSE.
- Add startVertex to the queue. // We use 0 as startVertex
- while (queue is not empty) do
  - Remove vertex, vi, from queue.
  - if (vi is not visited) then
    - Mark vi as visited and process it.
    - For every adjacent vertex, vj, to vi that is not visited
      - Add vj to queue
- Stop